UNIT 10 · STANDARD 6.G.2

Volume with Whole Number Edges

Lesson 10-1

Name: ______ Date: ______ Class: ______

Key Vocabulary

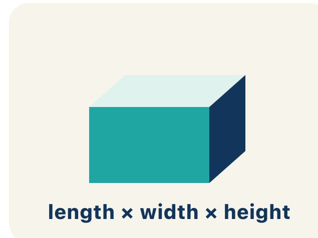
Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A box 3 × 2 × 4 holds 24 unit cubes, so V = 24 cubic units

Volume

How much space is inside a solid shape.



A cereal box or shoe box — it has length, width, and height

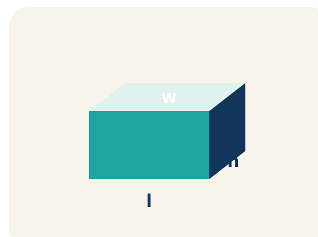
Rectangular prism

A solid box shape with six flat rectangle sides.

A tiny cube that is 1 in × 1 in × 1 in = 1 in³ (one cubic inch)

Cubic units

The units used to measure space inside, like cubic inches.



V = l × w × h: for a box 5 × 3 × 2, volume = 30 cubic units

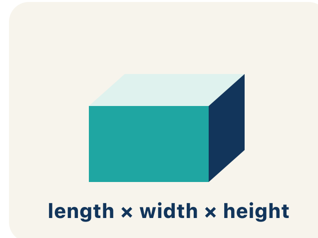
Length, width, height

How long, how wide, and how tall a box is.

A cross-shaped pattern of 6 rectangles folds into a rectangular box

Net

A flat shape that folds up into a solid.



A cube has 12 edges — 4 along the top, 4 along the bottom, 4 vertical

Edge

The line where two flat sides of a solid meet.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the volume of a rectangular prism with whole-number edges using length × width × height.

1️⃣ Notice

Your team is building time capsule containers to store class memories.

2️⃣ Key idea

I can find the volume of a rectangular prism with whole-number edges using length × width × height.

3️⃣ Words I'll use

Volume

Rectangular prism

Cubic units

Length, width, height

Net **Edge**

🤔 Think About It

- What three measurements do you need to find the volume of a box?
- What does volume tell us about a container?
- How is volume different from the area of one face?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: What is the volume of a rectangular prism with l = 7 in, w = 3 in, h = 4 in?

A. 84 in^3

B. 14 in^3

C. 84 in^2

D. 42 in^3

Step 1 $V = l \times w \times h = 7 \times 3 \times 4 = 84$ cubic inches.

✅ **Answer:** A. 84 in^3

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A time capsule box has a volume of 120 cm^3. Its length is 10 cm and width is 4 cm. What is its height?

A. 3 cm

B. 6 cm

C. 4 cm

D. 12 cm

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which unit is used for volume?

A. Cubic inches (in^3)

B. Square inches (in^2)

C. Inches (in)

D. Degrees (°)

Show your work:

____

____

____

Turn & Talk — Discussion Points

Level 1 support Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Your time capsule container is a rectangular prism. To find how much it can hold, why do you need three measurements (length, width, and height) instead of just one or two?

Level 1 support

Start here: Volume measures space in three directions, so you need length, width, and height.

💡 **Need a hint?**

Sentence starters (optional)

I need three measurements because volume fills space in ____ directions.

Necesito tres medidas porque el volumen llena el espacio en ____ direcciones.

Length and width alone only give the ____, not the volume.

El largo y el ancho solos solo dan el ____, no el volumen.

Word bank: volume | length | width | height | rectangular prism

Listen for: *Students explain that volume is three-dimensional, so all three edges (length, width, height) are needed; two measurements only give area of one face.*

Level 2

Could two time capsules have the same volume but different shapes? Justify with an example.

- Two capsules ____ have the same volume but different shapes because ____.
- For example, ____ × ____ × ____ and ____ × ____ × ____ both equal ____.

EXPLORE

Container C is $8 \times 3 \times 4$ and Container B is $5 \times 5 \times 2$. How did you count or multiply the unit cubes to find each volume, and which holds more?

Level 1 support

Start here: Multiply length × width × height to count all the unit cubes that fill the box.

💡 **Need a hint?**

Sentence starters (optional)

I found the volume by multiplying ___ × ___ × ___, which equals ___ cubic units.

Hallé el volumen multiplicando ___ × ___ × ___, que es ___ unidades cúbicas.

Container ___ holds more because its volume is ___.

El contenedor ___ contiene más porque su volumen es ___.

Word bank: volume | cubic units | length | width | height | rectangular prism

Listen for: *Students compute Container C = $8 \times 3 \times 4 = 96$ in^3 and Container B = $5 \times 5 \times 2 = 50$ in^3, and conclude C holds more by connecting volume to multiplying all three edges.*

Level 2

Container B (50 in^3) has a larger length and width than Container D ($10 \times 2 \times 3 = 60$ in^3), yet D holds more. Explain how that is possible.

- D holds more even though ___ because volume depends on ___.
- A bigger length or width does not always mean a bigger volume because ___.

CONNECT

A moving company's large box is $24 \times 18 \times 18$ in and its medium box is $18 \times 14 \times 12$ in. Talk through how you would find how many more cubic inches the large box holds.

Level 1 support

Start here: Find each volume with $V = l \times w \times h$, then subtract the smaller from the larger.

💡 **Need a hint?**

Sentence starters (optional)

The large box volume is ____ in³ because 24 × 18 × 18 = ____.

El volumen de la caja grande es ____ in³ porque 24 × 18 × 18 = ____.

The large box holds ____ more cubic inches because ____ − ____ = ____.

La caja grande contiene ____ pulgadas cúbicas más porque ____ − ____ = ____.

Word bank: volume | cubic units | rectangular prism | length | width | height

Listen for: *Students compute large = 7776 in³, medium = 3024 in³, then subtract to get 4752 in³ more, and reason that you must find each volume before comparing.*

Level 2

Why can't you just compare one dimension (like length) to decide which box holds more? Use the two boxes to justify.

- Comparing only length is not enough because ____.
- Even though the medium box is shorter in one edge, ____.

REFLECT

A classmate found the volume of a box and wrote the answer as 220 in^2. How would you correct the units, and how do you explain why volume uses cubic units?

Level 1 support

Start here: Volume measures three-dimensional space, so the units must be cubic (in^3), not square (in^2).

💡 **Need a hint?**

Sentence starters (optional)

The units should be ____ instead of in^2 because volume measures ____.

Las unidades deben ser ____ en lugar de in^2 porque el volumen mide ____.

Square units (in^2) are used for ____, but cubic units (in^3) are used for ____.

Las unidades cuadradas (in^2) se usan para ____, pero las unidades cúbicas (in^3) se usan para ____.

Word bank: volume | cubic units | area | rectangular prism | edge

Listen for: *Students change in^2 to in^3 and explain that volume fills 3D space (three edges multiplied), while square units describe flat area of one face.*

Level 2

If you multiplied inches × inches × inches, why does that produce cubic inches? Generalize the rule for any volume unit.

- Multiplying three lengths gives cubic units because ____.
- In general, the unit for volume is always ____ because ____.

Write About the Math

The Writing Revolution

I can explain my work using the words volume, rectangular prism, cubic units, and edge.

1. Kernel Sentence

subject + verb

Model: Volume is how much space is inside a solid shape.
Volumen es cuánto espacio hay dentro de una figura sólida.

Write a kernel sentence about volume. Use a subject and a verb.
Escribe una oración base sobre volumen. Usa un sujeto y un verbo.

2. Sentence Expansion

because · but · so

Kernel: Volume matters in math
Volumen importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Volume matters in math because ____.** *Volumen importa en matemáticas porque ____.*
but *pero*	**Volume matters in math, but ____.** *Volumen importa en matemáticas, pero ____.*
so *entonces*	**Volume matters in math, so ____.** *Volumen importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about volume.
Di un hecho verdadero sobre volume.

Volume ____.

Question
Pregunta

Ask a question about volume.
Haz una pregunta sobre volume.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about volume.
Muestra entusiasmo sobre volume.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with volume.
Dile a un compañero qué hacer con volume.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I need three measurements because volume fills space in ____ directions.
Necesito tres medidas porque el volumen llena el espacio en ____ direcciones.

Length and width alone only give the ____, not the volume.
El largo y el ancho solos solo dan el ____, no el volumen.

I found the volume by multiplying ____ × ____ × ____, which equals ____ cubic units.
Hallé el volumen multiplicando ____ × ____ × ____, que es ____ unidades cúbicas.

Try It

Solve on your own. Check the answer key when you are done.

1. A cube has edges of 4 units. What is its volume?

A. 64 cubic units

B. 16 cubic units

C. 12 cubic units

D. 48 cubic units

Show your work:

2. A box has volume 36 cubic units. Its base is 6 by 2. What is its height?

A. 3 units

B. 12 units

C. 6 units

D. 2 units

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?

Sentence starter: Option 1: ____ × ____ × ____. Option 2: ____ × ____ × ____. Option 3: ____ × ____ × ____. The ____ option is closest to a cube because ____. This matters because ____.

Show your work:

Reflect — Exit Ticket

A rectangular prism has l = 11 in, w = 5 in, h = 4 in. What is the volume?

A. 220 in^3

B. 55 in^3

C. 220 in^2

D. 200 in^3

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 3 cm *— $V = l \times w \times h \rightarrow 120 = 10 \times 4 \times h \rightarrow 120 = 40h \rightarrow h = 3$ cm.*
2. **You Try (notes):** A. Cubic inches (in^3) *— Volume measures 3D space, so it uses cubic units like in^3. Square units (in^2) are for area, and inches are for length.*
3. **Try It 1:** A. 64 cubic units *— $V = 4 \times 4 \times 4 = 64$ cubic units.*
4. **Try It 2:** A. 3 units *— Base area = 12; height = $36 \div 12 = 3$ units.*
5. **Exit Ticket:** A. 220 in^3 *— $V = 11 \times 5 \times 4 = 220$ cubic inches. Remember: volume uses cubic units (in^3).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Volume is how much space is inside a solid shape.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).